BOLUTIFE M. OMOLE

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EDUCATION

Massachusetts Institute of Technology Aug. 2025 – Expected May 2029
B.S. Electrical Engineering with Computing – GPA: 4.1/5.0 *Cambridge, MA*

- Relevant Coursework: Multivariable Calculus (18.02), Linear Algebra (18.06), Physics II (8.02), Electrical Circuits: Modeling and Design of Physical Systems (6.200), Discrete Math for Computer Science (6.1200), Programming in C and Assembly (6.1903), Dynamic System Modeling and Control Design (6.3100), U.S. Military Power (17.483)

EXPERIENCE

MIT Solar Electric Vehicle Team Aug. 2026 – Present
Electrical Engineer – Array Co-Lead *Cambridge, MA*

- Responsible for planning the design and implementation of the solar array for the electric vehicle.
- Developed and currently implementing a new method for sealing the array for improved wind resistance; replaced the bypass boards for better power distribution
- Researching new cells and designing the array layout for the 2028 vehicle
- Replaced Old Documentation with new documentation that is detailed and easy to read and follow

MIT Solar Electric Vehicle Team Sept. 2025 – Aug. 2026
Computer Programmer *Cambridge, MA*

- Worked with the Strategy Subteam to create an interactive dashboard with live route mapping and elevation profiling
- Included projected state-of-charge and solar insolation along the route to support real-time driver decision-making during the American Solar Challenge

WATABOT May 2026 – July 2026
Hardware Design Intern *Cambridge, MA*

- Designed a custom multi-function PCB in Altium consolidating 7 discrete control boards into a single integrated design, managing power regulation, dual-microcontroller interfacing, and 6-channel MOSFET actuator switching for a water-quality analysis instrument
- Took the design through DRC-clean layout to a complete, fabrication-ready output package (Gerbers, BOM, assembly drawings)

MIT Research Laboratory of Electronics Jan. 2026 – May 2026
Undergraduate Researcher *Cambridge, MA*

- Assisted MIT PhD Candidate Adam Gierlach to create a miniaturized PCB for a wireless gastric neural recording device (MiGUT)
- Developed embedded firmware for a Cortex-M3 microcontroller, implementing SPI communication translated from a Verilog testbench
- Performed PCB layout, bring-up, and debugging using Altium, validating ASIC communication via benchtop testing and oscilloscope measurements

SRT Communications Oct. 2024 – Aug. 2025
Technical Support Specialist *Minot, ND*

- Provided remote support for hardware, software, and network connectivity issues
- Accessed SRT routers, access points, and ONTs to diagnose and resolve technical problems

PROJECTS

Krebb One – HackMIT, Healthcare Track 2026

- Built the firmware and sensing hardware for a working prototype of a device that streams skin-contact and ambient temperature to a companion iPhone app over BLE at 1 Hz to visualize post-meal physiological response
- Designed the BLE packet contract with the software lead and implemented a quality heuristic to flag sensor and environmental noise

SKILLS

Programming Languages	Python, C, Assembly, HTML, CSS, JavaScript
Hardware & Engineering	PCB Design, Circuit Design, Soldering, Oscilloscope, Arduino, ESP32, Web Development
Tools & Platforms	Git, GitHub, VSCode, Altium Designer, Code Composer Studio
Hobbies	Basketball, Violin, Pickleball, Weightlifting